

10	Trigonometry	Notes
	Radian measure, including use for arc length and area of sector. The three basic trigonometric ratios of angles of any magnitude (in degrees or radians) and their graphs. Applications to simple problems in two or three dimensions (including angles between a line and a plane and between two planes). Use of the sine and cosine formulae. The identity $\cos^2 \theta + \sin^2 \theta = 1$. Use of the identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ The use of the basic addition formulae of trigonometry. Solution of simple trigonometric equations for a given interval.	The formulae $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ for a circle are expected to be known. To include the exact values for sine, cosine and tangent of 30°, 45°, 60° (and the radian equivalents), and the use of these to find the trigonometric ratios of related values such as 120°, 300°. General proofs of the sine and cosine formulae will not be required. The formulae are expected to be known. The area of a triangle in the form $\frac{1}{2}ab \sin C$ is expected to be known. $\cos^2 \theta + \sin^2 \theta = 1$ is expected to be known. $\tan \theta = \frac{\sin \theta}{\cos \theta}$ will be provided when needed. Formal proofs of basic formulae will not be required. Questions using the formulae for $\sin(A + B)$, $\cos(A + B)$, $\tan(A + B)$ may be set; the formulae will be provided, for example: $\sin(A + B) = \sin A \cos B + \cos A \sin B$ Long questions, explicitly involving excessive manipulation, will not be set. Students should be able to solve equations such as: $\sin\left(x - \frac{\pi}{2}\right) = \frac{3}{4}$ for $0 < x < 2\pi$, $\cos(x + 30^\circ) = \frac{1}{2}$ for $-180^\circ < x < 180^\circ$, $\tan 2x = 1$ for $90^\circ < x < 270^\circ$, $6 \cos^2 x^\circ + \sin x^\circ - 5 = 0$ for $0 \leq x < 360$, $\sin^2\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $-\pi \leq x < \pi$.